E/P **11 a** Sketch the graphs of $y = x^2(x - 1)(x + 1)$ and $y = \frac{1}{3}x^3 + 1$. **(5 marks)**

b Find the number of real solutions to the equation $3x^2(x - 1)(x + 1) = x^3 + 3$. **(1 mark)**

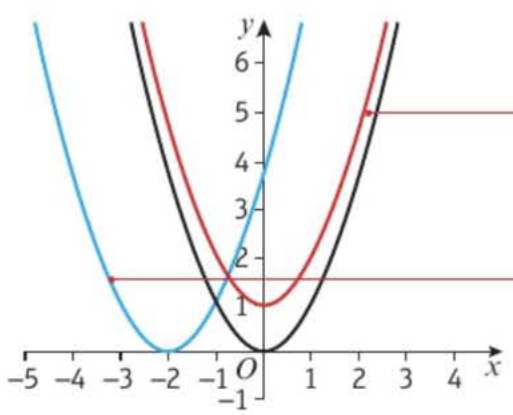
4.5 Translating graphs

You can transform the graph of a function by altering the function. Adding or subtracting a constant 'outside' the function translates a graph vertically.

- **The graph of $y = f(x) + a$ is a translation of the graph $y = f(x)$ by the vector $\begin{pmatrix} 0 \\ a \end{pmatrix}$.**

Adding or subtracting a constant 'inside' the function translates the graph horizontally.

- **The graph of $y = f(x + a)$ is a translation of the graph $y = f(x)$ by the vector $\begin{pmatrix} -a \\ 0 \end{pmatrix}$.**



$y = f(x) + 1$ is a translation $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, or 1 unit in the direction of the positive y-axis.

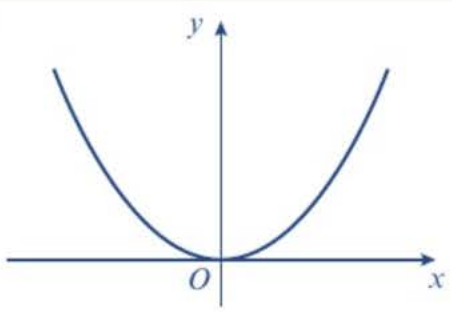
$y = f(x + 2)$ is a translation $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$, or 2 units in the direction of the negative x-axis.

Example 9

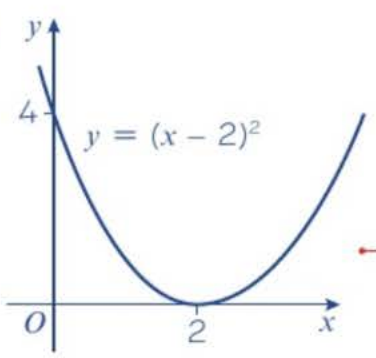
Sketch the graphs of:

a $y = x^2$ **b** $y = (x - 2)^2$ **c** $y = x^2 + 2$

a



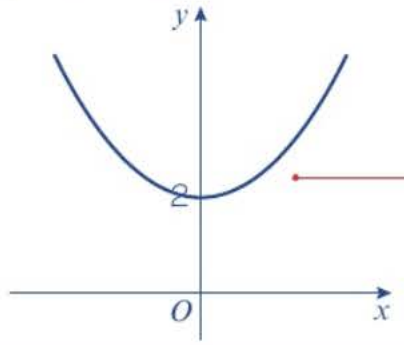
b



This is a translation by vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$.

Remember to mark on the intersections with the axes.

c $y = x^2 + 2$



This is a translation by vector $\begin{pmatrix}0\\2\end{pmatrix}$.

Remember to mark on the y-axis intersection.

Example 10

$f(x) = x^3$

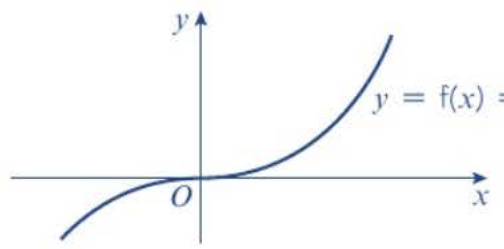
$g(x) = x(x - 2)$

Sketch the following graphs, indicating any points where the curves cross the axes:

a $y = f(x + 1)$

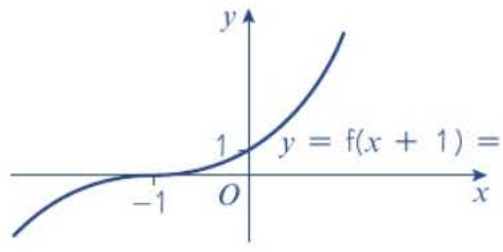
b $y = g(x + 1)$

a The graph of $f(x)$ is



First sketch $y = f(x)$.

So the graph of $y = f(x + 1)$ is



This is a translation of the graph of $y = f(x)$ by vector $\begin{pmatrix}-1\\0\end{pmatrix}$.

You could also write out the equation as $y = (x + 1)^3$ and sketch the graph directly.

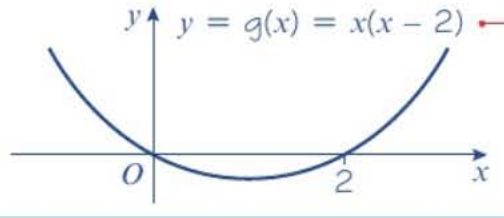
b $g(x) = x(x - 2)$

The curve is $y = x(x - 2)$

$0 = x(x - 2)$

Put $y = 0$ to find where the curve crosses the x-axis.

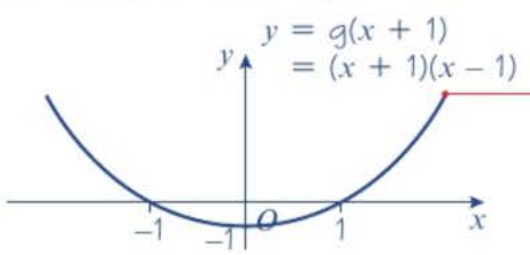
So $x = 0$ or $x = 2$



First sketch $g(x)$.

Online Explore translations of the graph of $y = x^3$ using GeoGebra.

So the graph of $y = g(x + 1)$ is



This is a translation of the graph of $y = g(x)$ by vector $\begin{pmatrix} -1 \\ 0 \end{pmatrix}$.

You could also write out the equation and sketch the graph directly:

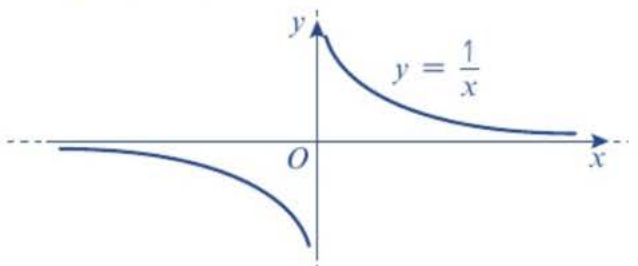
$y = g(x + 1)$
$= (x + 1)(x + 1 - 2)$
$= (x + 1)(x - 1)$

- **When you translate a function, any asymptotes are also translated.**

Example 11

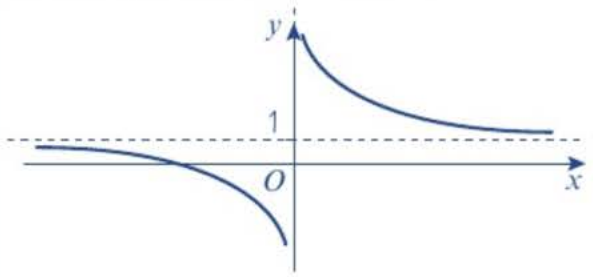
Given that $h(x) = \frac{1}{x}$, sketch the curve with equation $y = h(x) + 1$ and state the equations of any asymptotes and intersections with the axes.

The graph of $y = h(x)$ is



First sketch $y = h(x)$.

So the graph of $y = h(x) + 1$ is



The curve is translated by vector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ so the asymptote is translated by the same vector.

The curve crosses the x-axis once.

$$y = h(x) + 1 = \frac{1}{x} + 1$$

$$0 = \frac{1}{x} + 1$$

$$-1 = \frac{1}{x}$$

$$x = -1$$

Put $y = 0$ to find where the curve crosses the x-axis.

So the curve intersects the x-axis at $(-1, 0)$.

The horizontal asymptote is $y = 1$.

The vertical asymptote is $x = 0$.

Remember to write down the equation of the vertical asymptote as well. It is the y-axis so it has equation $x = 0$.

Exercise 4E

1 Apply the following transformations to the curves with equations $y = f(x)$ where:

i $f(x) = x^2$ **ii** $f(x) = x^3$ **iii** $f(x) = \frac{1}{x}$

In each case state the coordinates of points where the curves cross the axes and in **iii** state the equations of the asymptotes.

a $f(x + 2)$ **b** $f(x) + 2$ **c** $f(x - 1)$

d $f(x) - 1$ **e** $f(x) - 3$ **f** $f(x - 3)$

2 a Sketch the curve $y = f(x)$ where $f(x) = (x - 1)(x + 2)$.

b On separate diagrams sketch the graphs of **i** $y = f(x + 2)$ **ii** $y = f(x) + 2$.

c Find the equations of the curves $y = f(x + 2)$ and $y = f(x) + 2$, in terms of x, and use these equations to find the coordinates of the points where your graphs in part **b** cross the y-axis.

3 a Sketch the graph of $y = f(x)$ where $f(x) = x^2(1 - x)$.

b Sketch the curve with equation $y = f(x + 1)$.

c By finding the equation $f(x + 1)$ in terms of x, find the coordinates of the point in part **b** where the curve crosses the y-axis.

4 a Sketch the graph of $y = f(x)$ where $f(x) = x(x - 2)^2$.

b Sketch the curves with equations $y = f(x) + 2$ and $y = f(x + 2)$.

c Find the coordinates of the points where the graph of $y = f(x + 2)$ crosses the axes.

5 a Sketch the graph of $y = f(x)$ where $f(x) = x(x - 4)$.

b Sketch the curves with equations $y = f(x + 2)$ and $y = f(x) + 4$.

c Find the equations of the curves in part **b** in terms of x and hence find the coordinates of the points where the curves cross the axes.

6 a Sketch the graph of $y = f(x)$ where $f(x) = x^2(x - 1)(x - 2)$.

b Sketch the curves with equations $y = f(x + 2)$ and $y = f(x) - 1$.

(E) **7** The point $P(4, -1)$ lies on the curve with equation $y = f(x)$.

a State the coordinates that point P is transformed to on the curve with equation $y = f(x - 2)$. **(1 mark)**

b State the coordinates that point P is transformed to on the curve with equation $y = f(x) + 3$. **(1 mark)**

(E/P) **8** The graph of $y = f(x)$ where $f(x) = \frac{1}{x}$ is translated so that the asymptotes are at $x = 4$ and $y = 0$. Write down the equation for the transformed function in the form $y = \frac{1}{x + a}$ **(3 marks)**

(P) **9 a** Sketch the graph of $y = x^3 - 5x^2 + 6x$, marking clearly the points of intersection with the axes.

b Hence sketch $y = (x - 2)^3 - 5(x - 2)^2 + 6(x - 2)$.

(P) **10** **a** Sketch the graph of $y = x^2(x - 3)(x + 2)$, marking clearly the points of intersection with the axes.

b Hence sketch $y = (x + 2)^2(x - 1)(x + 4)$.

(E/P) **11** **a** Sketch the graph of $y = x^3 + 4x^2 + 4x$. **(6 marks)**

b The point with coordinates $(-1, 0)$ lies on the curve with equation $y = (x + a)^3 + 4(x + a)^2 + 4(x + a)$ where a is a constant. Find the two possible values of a. **(3 marks)**

Problem-solving

Look at your sketch and picture the curve sliding to the left or right.

(E/P) **12** **a** Sketch the graph of $y = x(x + 1)(x + 3)^2$. **(4 marks)**

b Find the possible values of b such that the point $(2, 0)$ lies on the curve with equation $y = (x + b)(x + b + 1)(x + b + 3)^2$. **(3 marks)**

Challenge

1 Sketch the graph of $y = (x - 3)^3 + 2$ and determine the coordinates of the point of inflection. → Section 12.9

2 The point $Q(-5, -7)$ lies on the curve with equation $y = f(x)$.

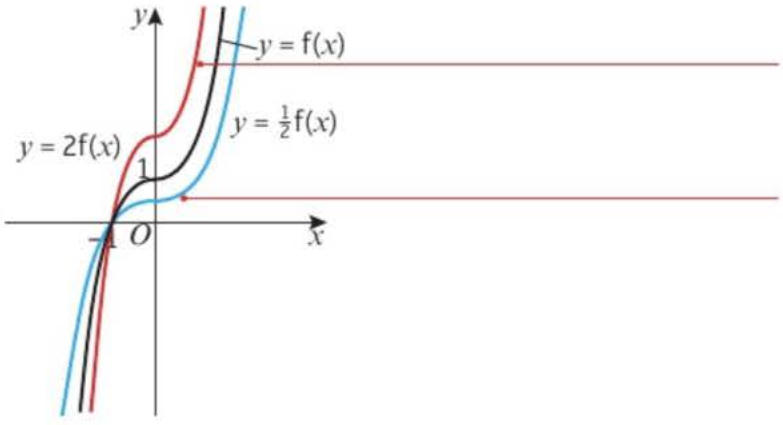
a State the coordinates that point Q is transformed to on the curve with equation $y = f(x + 2) - 5$.

b The coordinates of the point Q on a transformed curve are $(-3, -6)$. Write down the transformation in the form $y = f(x + a) - b$.

4.6 Stretching graphs

Multiplying by a constant 'outside' the function stretches the graph vertically.

- **The graph of $y = af(x)$ is a stretch of the graph $y = f(x)$ by a scale factor of a in the vertical direction.**

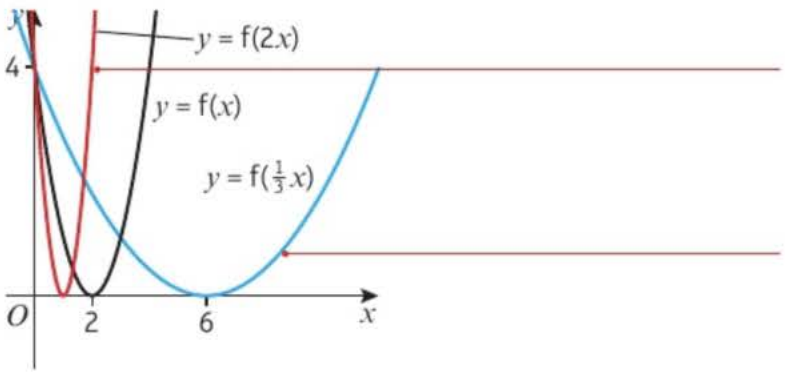


$2f(x)$ is a stretch with scale factor 2 in the y-direction. All y-coordinates are doubled.

$\frac{1}{2}f(x)$ is a stretch with scale factor $\frac{1}{2}$ in the y-direction. All y-coordinates are halved.

Multiplying by a constant 'inside' the function stretches the graph horizontally.

- **The graph of $y = f(ax)$ is a stretch of the graph $y = f(x)$ by a scale factor of $\frac{1}{a}$ in the horizontal direction.**



$y = f(2x)$ is a stretch with scale factor $\frac{1}{2}$ in the x-direction. All x-coordinates are halved.

$y = f(\frac{1}{3}x)$ is a stretch with scale factor 3 in the x-direction. All x-coordinates are tripled.

Example 12

Given that $f(x) = 9 - x^2$, sketch the curves with equations:

a $y = f(2x)$ **b** $y = 2f(x)$

a $f(x) = 9 - x^2$

So $f(x) = (3 - x)(3 + x)$

You can factorise the expression.

The curve is $y = (3 - x)(3 + x)$

$0 = (3 - x)(3 + x)$

Put $y = 0$ to find where the curve crosses the x-axis.

So $x = 3$ or $x = -3$

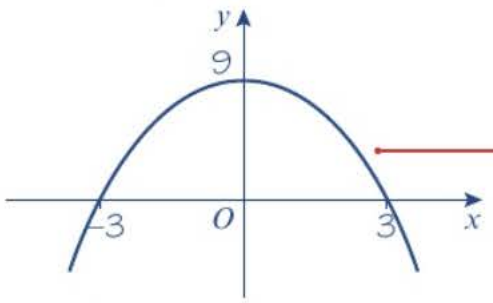
So the curve crosses the x-axis at (3, 0) and (−3, 0).

When $x = 0$, $y = 3 \times 3 = 9$

Put $x = 0$ to find where the curve crosses the y-axis.

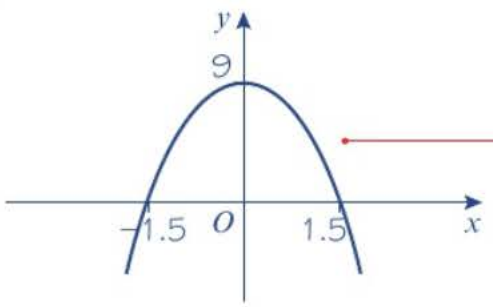
So the curve crosses the y-axis at (0, 9).

The curve $y = f(x)$ is



First sketch $y = f(x)$.

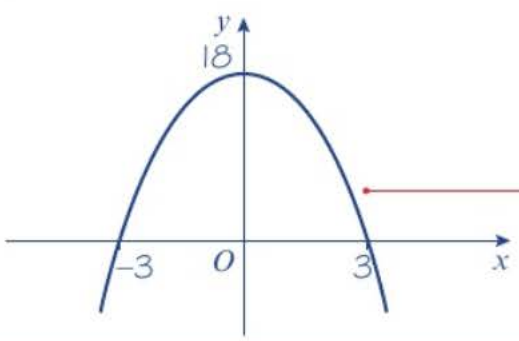
$y = f(2x)$ so the curve is



$y = f(ax)$ where $a = 2$ so it is a horizontal stretch with scale factor $\frac{1}{2}$.

Check: The curve is $y = f(2x)$.
So $y = (3 - 2x)(3 + 2x)$.
When $y = 0$, $x = -1.5$ or $x = 1.5$.
So the curve crosses the x-axis at (−1.5, 0) and (1.5, 0).
When $x = 0$, $y = 9$.
So the curve crosses the y-axis at (0, 9).

b $y = 2f(x)$ so the curve is



$y = af(x)$ where $a = 2$ so it is a vertical stretch with scale factor 2.

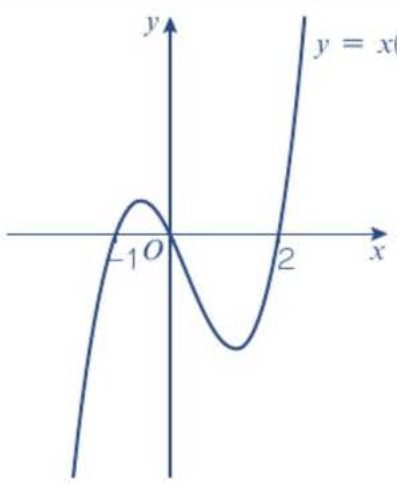
Check: The curve is $y = 2f(x)$.
So $y = 2(3 - x)(3 + x)$.
When $y = 0$, $x = 3$ or $x = -3$.
So the curve crosses the x-axis at (−3, 0) and (3, 0).
When $x = 0$, $y = 2 \times 9 = 18$.
So the curve crosses the y-axis at (0, 18).

Example 13

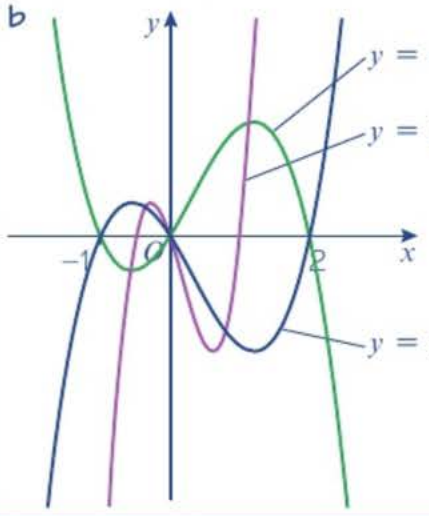
a Sketch the curve with equation $y = x(x - 2)(x + 1)$.

b On the same axes, sketch the curves $y = 2x(2x - 2)(2x + 1)$ and $y = -x(x - 2)(x + 1)$.

a



b



Online Explore stretches of the graph of $y = x(x - 2)(x + 1)$ using GeoGebra.

$y = -x(x - 2)(x + 1)$ is a stretch with scale factor -1 in the y-direction. Notice that this stretch has the effect of reflecting the curve in the x-axis.

$y = 2x(2x - 2)(2x + 1)$ is a stretch with scale factor $\frac{1}{2}$ in the x-direction.

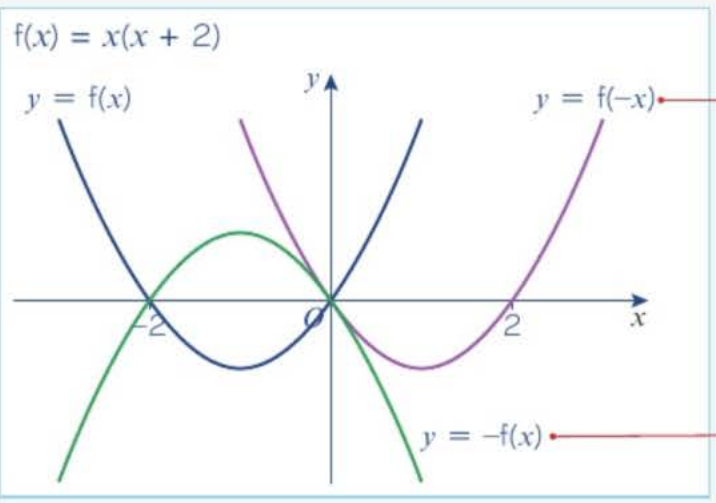
Problem-solving

You need to work out the relationship between each new function and the original function.
If $x(x - 2)(x + 1) = f(x)$ then
$2x(2x - 2)(2x + 1) = f(2x)$, and
$-x(x - 2)(x + 1) = -f(x)$.

- **The graph of $y = -f(x)$ is a reflection of the graph of $y = f(x)$ in the x-axis.**
- **The graph of $y = f(-x)$ is a reflection of the graph of $y = f(x)$ in the y-axis.**

Example 14

On the same axes sketch the graphs of $y = f(x)$, $y = f(-x)$ and $y = -f(x)$ where $f(x) = x(x + 2)$.



$y = f(-x)$ is $y = (-x)(-x + 2)$ which is $y = x^2 - 2x$ or $y = x(x - 2)$ and this is a reflection of the original curve in the y-axis.

Alternatively multiply each x-coordinate by -1 and leave the y-coordinates unchanged.

This is the same as a stretch parallel to the x-axis scale factor -1.

$y = -f(x)$ is $y = -x(x + 2)$ and this is a reflection of the original curve in the x-axis.

Alternatively multiply each y-coordinate by -1 and leave the x-coordinates unchanged.

This is the same as a stretch parallel to the y-axis scale factor -1.

Exercise 4F

1 Apply the following transformations to the curves with equations $y = f(x)$ where:

i $f(x) = x^2$ **ii** $f(x) = x^3$ **iii** $f(x) = \frac{1}{x}$

In each case show both $f(x)$ and the transformation on the same diagram.

a $f(2x)$ **b** $f(-x)$ **c** $f(\frac{1}{2}x)$ **d** $f(4x)$ **e** $f(\frac{1}{4}x)$

f $2f(x)$ **g** $-f(x)$ **h** $4f(x)$ **i** $\frac{1}{2}f(x)$ **j** $\frac{1}{4}f(x)$

2 a Sketch the curve with equation $y = f(x)$ where $f(x) = x^2 - 4$.

b Sketch the graphs of $y = f(4x)$, $\frac{1}{3}y = f(x)$, $y = f(-x)$ and $y = -f(x)$.

Hint For part **b**, rearrange the second equation into the form $y = 3f(x)$.

3 a Sketch the curve with equation $y = f(x)$ where $f(x) = (x - 2)(x + 2)x$.

b Sketch the graphs of $y = f(\frac{1}{2}x)$, $y = f(2x)$ and $y = -f(x)$.

(P) **4 a** Sketch the curve with equation $y = x^2(x - 3)$.

b On the same axes, sketch the curves with equations:

i $y = (2x)^2(2x - 3)$ **ii** $y = -x^2(x - 3)$

Problem-solving

Let $f(x) = x^2(x - 3)$ and try to write each of the equations in part **b** in terms of $f(x)$.

5 a Sketch the curve $y = x^2 + 3x - 4$.

b On the same axes, sketch the graph of $5y = x^2 + 3x - 4$.

6 a Sketch the graph of $y = x^2(x - 2)^2$.

b On the same axes, sketch the graph of $3y = -x^2(x - 2)^2$.

(E) **7** The point $P(2, -3)$ lies on the curve with equation $y = f(x)$.

a State the coordinates that point P is transformed to on the curve with equation $y = f(2x)$. **(1 mark)**

b State the coordinates that point P is transformed to on the curve with equation $y = 4f(x)$. **(1 mark)**

(E) **8** The point $Q(-2, 8)$ lies on the curve with equation $y = f(x)$.

State the coordinates that point Q is transformed to on the curve with equation $y = f(\frac{1}{2}x)$. **(1 mark)**

(E/P) **9 a** Sketch the graph of $y = (x - 2)(x - 3)^2$. **(4 marks)**

b The graph of $y = (ax - 2)(ax - 3)^2$ passes through the point $(1, 0)$.
Find two possible values for a. **(3 marks)**

Challenge

1 The point $R(4, -6)$ lies on the curve with equation $y = f(x)$. State the coordinates that point R is transformed to on the curve with equation $y = \frac{1}{3}f(2x)$.

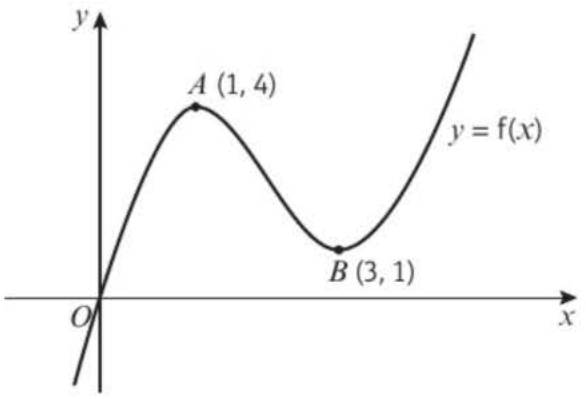
2 The point $S(-4, 7)$ is transformed to a point $S'(-8, 1.75)$. Write down the transformation in the form $y = af(bx)$.

4.7 Transforming functions

You can apply transformations to unfamiliar functions by considering how specific points and features are transformed.

Example 15

The following diagram shows a sketch of the curve f(x) which passes through the origin.
The points $A(1, 4)$ and $B(3, 1)$ also lie on the curve.



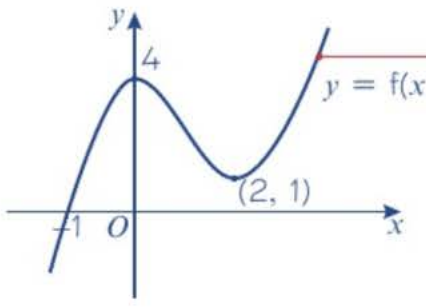
Sketch the following:

a $y = f(x + 1)$ **b** $y = f(x - 1)$ **c** $y = f(x) - 4$

d $2y = f(x)$ **e** $y - 1 = f(x)$

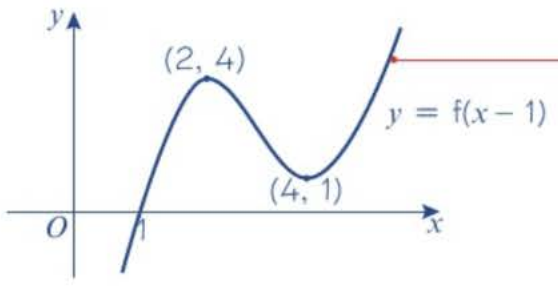
In each case you should show the positions of the images of the points O, A and B.

a f(x + 1)



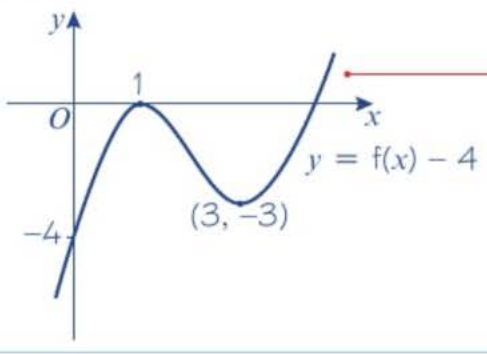
Translate f(x) 1 unit in the direction of the negative x-axis.

b f(x − 1)



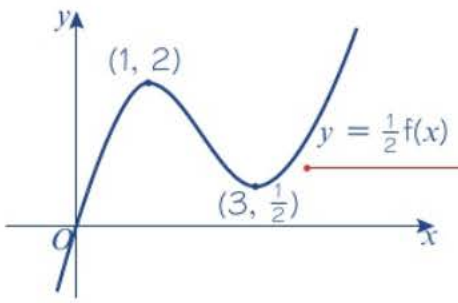
Translate f(x) 1 unit in the direction of the positive x-axis.

c f(x) − 4



Translate f(x) 4 units in the direction of the negative y-axis.

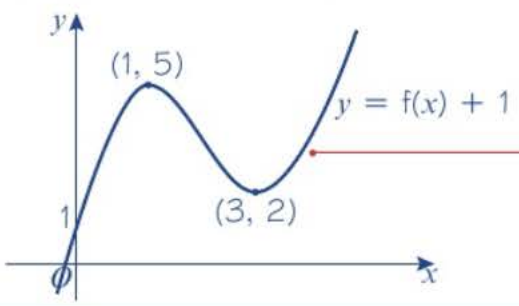
d $2y = f(x)$ so $y = \frac{1}{2}f(x)$



Rearrange in the form $y = \ldots$

Stretch $f(x)$ by scale factor $\frac{1}{2}$ in the y-direction.

e $y - 1 = f(x)$ so $y = f(x) + 1$

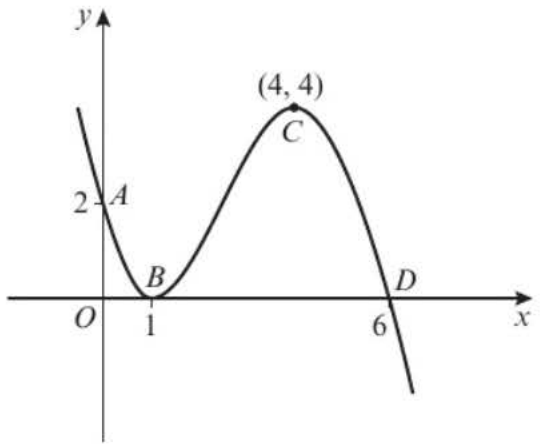


Rearrange in the form $y = \ldots$

Translate $f(x)$ 1 unit in the direction of the positive y-axis.

Exercise 4G

1 The following diagram shows a sketch of the curve with equation $y = f(x)$. The points $A(0, 2)$, $B(1, 0)$, $C(4, 4)$ and $D(6, 0)$ lie on the curve.



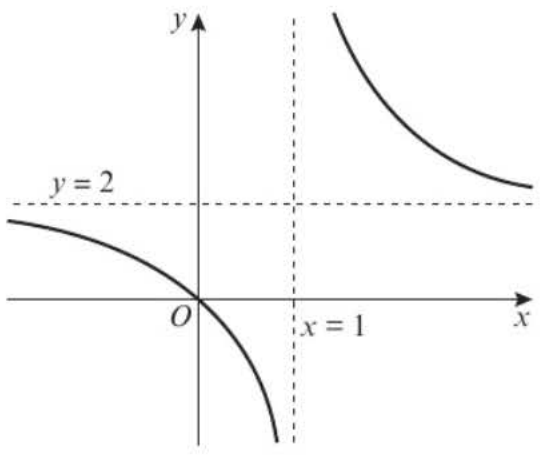
Sketch the following graphs and give the coordinates of the points, A, B, C and D after each transformation:

a $f(x + 1)$ **b** $f(x) - 4$ **c** $f(x + 4)$

d $f(2x)$ **e** $3f(x)$ **f** $f(\frac{1}{2}x)$

g $\frac{1}{2}f(x)$ **h** $f(-x)$

2 The curve $y = f(x)$ passes through the origin and has horizontal asymptote $y = 2$ and vertical asymptote $x = 1$, as shown in the diagram.



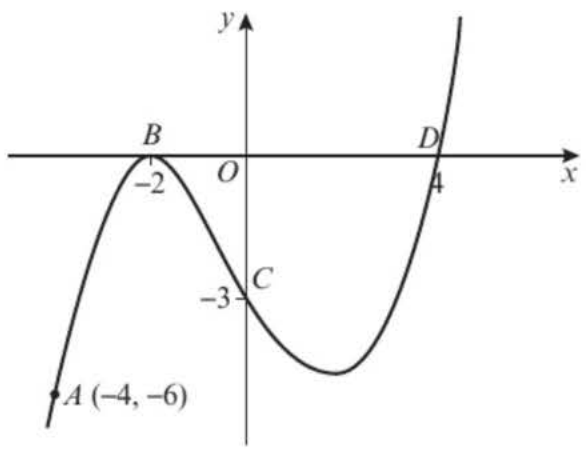
Sketch the following graphs. Give the equations of any asymptotes and give the coordinates of intersections with the axes after each transformation.

a $f(x) + 2$ **b** $f(x + 1)$ **c** $2f(x)$

d $f(x) - 2$ **e** $f(2x)$ **f** $f(\frac{1}{2}x)$

g $\frac{1}{2}f(x)$ **h** $-f(x)$

3 The curve with equation $y = f(x)$ passes through the points $A(-4, -6)$, $B(-2, 0)$, $C(0, -3)$ and $D(4, 0)$ as shown in the diagram.



Sketch the following and give the coordinates of the points A, B, C and D after each transformation.

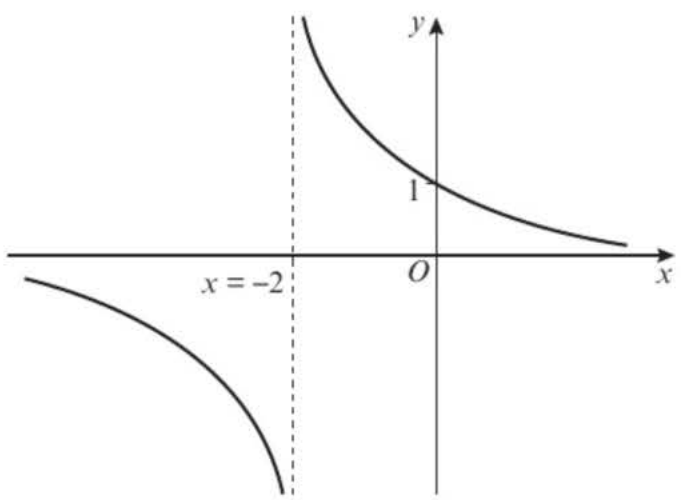
a $f(x - 2)$ **b** $f(x) + 6$ **c** $f(2x)$

d $f(x + 4)$ **e** $f(x) + 3$ **f** $3f(x)$

g $\frac{1}{3}f(x)$ **h** $f(\frac{1}{4}x)$ **i** $-f(x)$

j $f(-x)$

4 A sketch of the curve $y = f(x)$ is shown in the diagram. The curve has a vertical asymptote with equation $x = -2$ and a horizontal asymptote with equation $y = 0$. The curve crosses the y-axis at $(0, 1)$.



a Sketch, on separate diagrams, the graphs of:

i $2f(x)$ **ii** $f(2x)$ **iii** $f(x - 2)$

iv $f(x) - 1$ **v** $f(-x)$ **vi** $-f(x)$

In each case state the equations of any asymptotes and, if possible, points where the curve cuts the axes.

b Suggest a possible equation for $f(x)$.

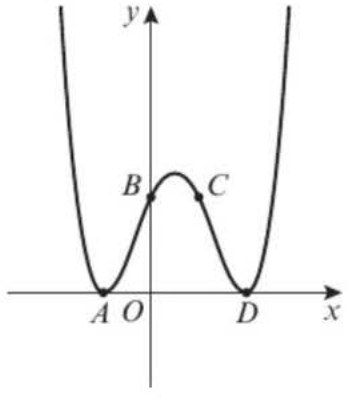
E/P **5** The point $P(2, 1)$ lies on the graph with equation $y = f(x)$.

a On the graph of $y = f(ax)$, the point P is mapped to the point $Q(4, 1)$. Determine the value of a. **(1 mark)**

b Write down the coordinates of the point to which P maps under each transformation

i $f(x - 4)$ **ii** $3f(x)$ **iii** $\frac{1}{2}f(x) - 4$ **(3 marks)**

P **6** The diagram shows a sketch of a curve with equation $y = f(x)$. The points $A(-1, 0)$, $B(0, 2)$, $C(1, 2)$ and $D(2, 0)$ lie on the curve.



Sketch the following graphs and give the coordinates of the points A, B, C and D after each transformation:

a $y + 2 = f(x)$ **b** $\frac{1}{2}y = f(x)$

c $y - 3 = f(x)$ **d** $3y = f(x)$

e $2y - 1 = f(x)$

Problem-solving

Rearrange each equation into the form $y = \ldots$

Mixed exercise 4

1 **a** On the same axes sketch the graphs of $y = x^2(x - 2)$ and $y = 2x - x^2$.

b By solving a suitable equation find the points of intersection of the two graphs.

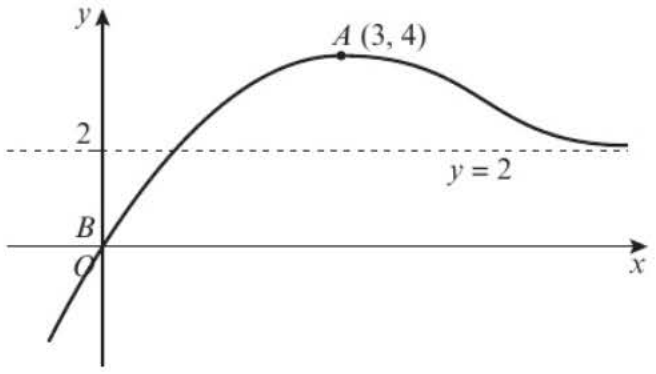
(P) **2** **a** On the same axes sketch the curves with equations $y = \frac{6}{x}$ and $y = 1 + x$.

b The curves intersect at the points A and B. Find the coordinates of A and B.

c The curve C with equation $y = x^2 + px + q$, where p and q are integers, passes through A and B. Find the values of p and q.

d Add C to your sketch.

3 The diagram shows a sketch of the curve $y = f(x)$. The point $B(0, 0)$ lies on the curve and the point $A(3, 4)$ is a maximum point. The line $y = 2$ is an asymptote.



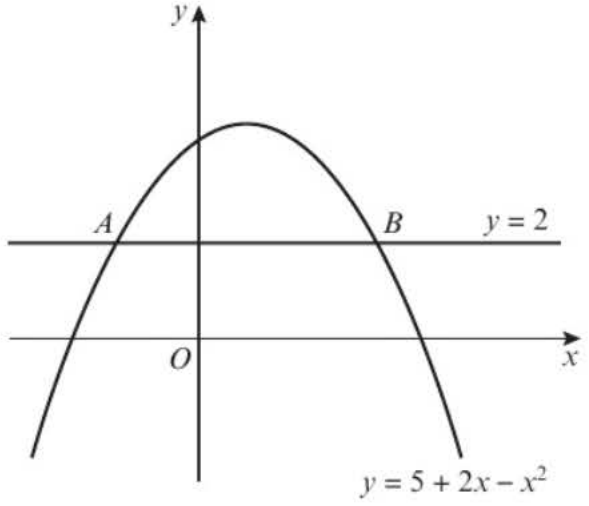
Sketch the following and in each case give the coordinates of the new positions of A and B and state the equation of the asymptote:

a $f(2x)$ **b** $\frac{1}{2}f(x)$ **c** $f(x) - 2$

d $f(x + 3)$ **e** $f(x - 3)$ **f** $f(x) + 1$

(E) **4** The diagram shows the curve with equation $y = 5 + 2x - x^2$ and the line with equation $y = 2$. The curve and the line intersect at the points A and B.

Find the x-coordinates of A and B. **(4 marks)**



(E/P) **5** $f(x) = x^2(x - 1)(x - 3)$.

a Sketch the graph of $y = f(x)$. **(2 marks)**

b On the same axes, draw the line $y = 2 - x$. **(2 marks)**

c State the number of real solutions to the equation $x^2(x - 1)(x - 3) = 2 - x$. **(1 mark)**

d Write down the coordinates of the point where the graph with equation $y = f(x) + 2$ crosses the y-axis. **(1 mark)**

(E) **6** The figure shows a sketch of the curve with equation $y = f(x)$.

On separate axes sketch the curves with equations:

a $y = f(-x)$ **(2 marks)**

b $y = -f(x)$ **(2 marks)**

Mark on each sketch the x-coordinate of any point, or points, where the curve touches or crosses the x-axis.

